HDDet: A More Common Heading Direction Detector for Remote Sensing and Arbitrary Viewing Angle Images

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***Abstract*— Object heading detection (OHD) offers potential for research in the control and traffic analysis sectors. Contemporary methodologies in OHD grapple with a set of distinct limitations: a constrained range of detectable object types, a discernible drop in accuracy for oriented bounding box (OBB) predictions influenced by heading estimations, and the inherent limitations associated with the exclusive perspective of bird's-eye view imagery. This article introduces heading direction detector (HDDet), an advanced method devised for detecting the OBB of objects with heading direction from various viewpoints. It sequentially delineates the circular annotation method (CAM), multidimensional angle encoding (MDAE), and the CosWeight strategy. CAM initially couples OBB data with heading details for accurate target delineation. MDAE follows, optimizing angle encoding to boost the model's training process. The culmination of this approach is CosWeight, which integrates the Rotated-intersection over union (IoU) and heading information into the horizontal box's loss, thereby enhancing the precision of heading predictions. Rigorous testing across diverse datasets, including the SJTU-L dataset where the heading accuracy increased from 64.41% to 94.82% over OHDet, validates HDDet's enhanced capabilities, surpassing existing methodologies in both OBB detection and heading accuracy, and marking a significant advancement over the state-of-the-art OHDet. Additionally, this article presents an engineering vehicle dataset, which is conducive to multiperspective OHD research.**

***Index Terms*— Data augmentation, deep learning, intersection over union (IoU) loss, object heading detection (OHD), remote sensing.**

I. Introduction

ORIENTED object detection, a significant and evolving branch within the object detection domain, has gained considerable attention in recent years. This method, superior in detecting objects with various orientations and dense arrangements by oriented bounding box (OBB), significantly outperforms traditional horizontal bounding boxes (HBBs). Initially employed for detecting non-parallel text [1], [2], this method has been adeptly applied to remote sensing imagery for identifying small, densely packed, and rotated objects [3], [4]. Recently, notable OBB detectors like SCRDet [5], MFF module [6], R^3Det [7], and ReDet [8] have marked significant advancements in this field.



Fig. 1. Comparison of detection results with (a) HBB, (b) OBB, and (c) object heading detector.

However, in surveillance and control applications, OBB detectors, despite their advantages over HBB detectors in depicting object locations and areas, do not fully convey the heading information of objects. This limitation is a significant hurdle in achieving precise monitoring and control. As Fig. 1 illustrates, while OBB detectors accurately pinpoint the location of trucks, they fall short in revealing the vehicles' actual state—for instance, discerning if a truck is moving forward or backward, and establishing the relationship between the orientation of the truck and its direction of motion. A comprehensive understanding of a vehicle's state, as demonstrated in subfigure (c), is feasible only when its heading is explicitly indicated.

In the quest to identify the heading information of objects, researchers have proposed new methods for object heading detection (OHD). A notable example is OPD-Net [9], which pioneered a network aimed at estimating the heading by pinpointing ship prows through advanced regression methods. Yet, while OPD-Net demonstrates exceptional proficiency in detecting ship prows, its specialized focus may limit its versatility in handling other object categories. Subsequently, the OHDet [10] is proposed by simply adding a four-category approach to the baseline of CSL [11], to further classify which

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Data is available on-line at https://github.com/G-virus/YOLOv6-HDDet.

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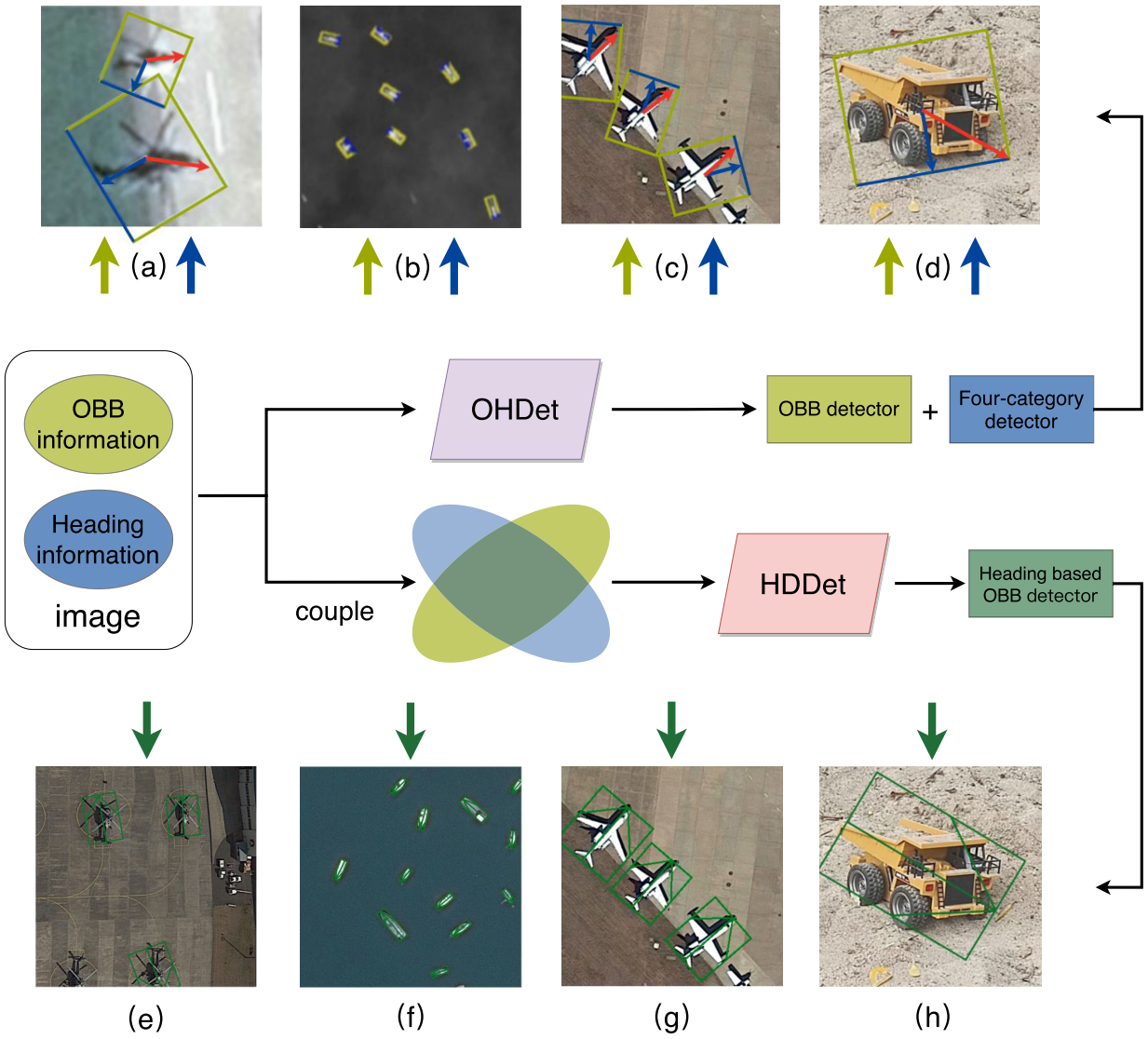


Fig. 2. Comparison of OHDet and HDDet is illustrated in the figure. (a)–(d) Predictive results using an additional four-category paradigm. (e)–(h) Predictive results when the information of the OBB and heading is coupled. In (a)–(d), light green OBBs and blue lines represent the detected results by OBB and four-category detectors, respectively. However, the red arrows indicate the actual heading of the objects. In (e)–(h), dark green OBBs with arrows show the detected results achieved by coupling the information of OBB and heading.

sides of OBB the head is located in. Benefiting from the precise detection performance of CSL, OHDet can also accurately detect the heading of different remote sensing objects. Moreover, a new dataset, annotated with heading direction markers [10], named OHD-SJTU, proves the effectiveness of this method.

Nonetheless, the additional four-category paradigm, while innovative, does not fully associate and couple the object heading with OBB information, leading to notable limitations. As evident in Fig. 2(a) and (b), even when OBB detection is precise, the accuracy of heading detection can falter if the heading data is ambiguous. In contrast, Fig. 2(e) and (f) illustrates that coupling this information markedly improves the accuracy in determining the heading. Furthermore, as depicted in Fig. 2(c) and (d), there are instances where the orientation of the OBB does not necessarily correspond with the heading. In such situations, none of the OBB edges can precisely represent the heading, culminating in inaccurate heading predictions. However, this challenge is effectively mitigated by coupling the OBB and heading information, as shown in Fig. 2(g) and (h). Moreover, this four-category approach, while offering some advantages, can erode the precision of detection. As mentioned in OHDet [10], the average precision (AP) with OHDet on the OHD-SJTU-L dataset plummets from a robust 65.92% in OBB detection to a modest 38.04% in OHD detection.

To correctly couple the information of OBB and heading, this work introduces the heading direction detector (HDDet) method. First, a new annotation method named circular annotation method (CAM) is proposed. CAM couples the information of OBB and heading by using the angle of the object's heading direction as the rotational angle of the bounding box. That is to say, CAM extends the angle information of OBB by converting the angle information into the heading information of the

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objects within a circle. This simultaneous angle annotation of both the rotational box and heading direction significantly enhances the synergy between these detections. Following this, the FullCSL angle representation is designed to encode and predict the rotational angle of the OBB, as well as the heading direction, using 360 angle classifications. To expedite model training, FullCSL is enhanced with a novel multidimensional angle encoding (MDAE) technique, which greatly reduces the encoding length of the angle and speeds up the training, making it more friendly to the training of lightweight models. Furthermore, to integrate the intersection over union (IoU) [12] loss of the OBB and heading prediction into the bounding box training process, and to further enhance the performance of OHD, a CosWeight is developed. The effectiveness of HDDet is demonstrated through testing with datasets such as DOTA and our custom-engineered vehicle dataset. The experimental results verify that HDDet not only precisely identifies object heading but also attains a mean AP (mAP) that rivals the state-of-the-art detectors in OBB detection tasks. The contributions of this work are summarized as follows.

1) The CAM is proposed to couple the information of OBB and heading.

2) The MDAE technique is designed to not only reduce the complexity of the model's head layer but also accelerate its training process.

3) CosWeight is designed as a strategy refining the conventional IoU loss, by infusing it with a corrected weight for heading detection.

4) An engineering vehicle dataset for OHD with arbitrary viewing angles is designed. According to the detection results of this dataset, the performance of our HDDet is verified.

The subsequent sections of this article are methodically structured. Section II delves into the pertinent literature, focusing on existing bounding box and IoU representation techniques. Section III sequentially describes CAM, FullCSL, MDAE, and CosWeight. Section IV presents the experiment results of our HDDet, in comparison with the state-of-the-art methods. Section V encapsulates the study with concluding remarks.

II. Related Work

A. Object Detection With Different Bounding Boxes

1) Detection With HBB: The realm of traditional object detection has witnessed a rapid evolution, leading to its prevalent application across diverse sectors. Prototypes like R-CNN [13], and its advanced iterations, Fast R-CNN [14] and Faster R-CNN [15], as well as FPN [16], epitomize the efficacy of two-stage prediction models. In contrast, single-stage models such as SSD [17], the YOLO series [18], [19], [20], [21], [22], [23], [24], and RetinaNet [25] are renowned for their swift processing capabilities, ideal for real-time applications, albeit with a slight compromise in accuracy. Rooted in HBB annotations, these foundational models encounter notable challenges in accurately detecting objects with varying orientations, often resulting in substantial overlap dilemmas in specific contexts.

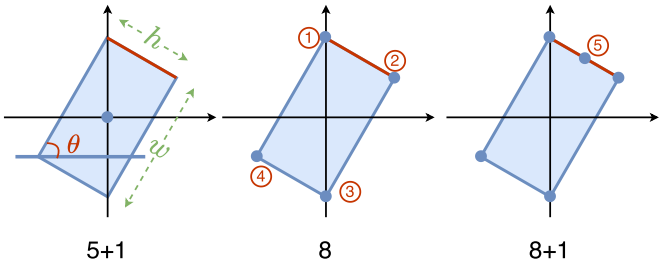


Fig. 3. Transformation of prevalent OBB annotation approaches into OHD angle annotation approaches. The "5+1" method employs OpenCV's five-parameter formula to pinpoint the bounding box's edge where the head is located. The red edge represents the heading of object. "8" method utilizes a consistent vertex order to determine the heading's edge, which presents heading direction using point sequencing. The edge represented by the first two vertices are the heading of object. "8+1" builds upon the "8" approach by adding the midpoint of the heading direction for enhanced accuracy.

2) Detection With OBB: The pursuit of detecting objects of arbitrary orientation in remote sensing has emerged as a focal area of contemporary research. Addressing this challenge in HBB necessitates the addition of an angular dimension. To this end, five-parameter and eight-parameter annotation methods are proposed, respectively. The five-parameter approach, which directly adds a rotational angle based on the HBB annotation, grapples with issues such as the periodicity of angular (PoA) and exchangeability of edges (EoEs). In contrast, the eight-parameter method, delineating the bounding box through the coordinates of four points (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), resolves these specific challenges but encounters boundary discontinuity problems. This issue adversely impacts the accuracy in IoU computations and optimization of loss functions. Leading models that address these complexities include RoI Transformer [26], R^3det [7], RRD [27], and CSL [11].

3) Object Heading Detection: In the era preceding the dominance of deep learning frameworks, Liu and Mattyus [28] were trailblazers in detecting vehicle heading directions, innovatively employing a sliding window strategy in tandem with binary classification. This approach marked a seminal step in the field, providing a fundamental yet initial approximation of the vehicle's head position. With the advent of deep learning, the methodology evolved, enriching the OBB technique by adding a crucial dimension to precisely arrange the four vertices of the prediction box. A prominent advancement in this domain is the Gliding Vertex [29], which meticulously arranges the vertices in the eight-parameter representation, thereby finetuning the order of the predicted points, as illustrated in Fig. 3 under "8." This methodical sequencing not only solidifies the consistency in identifying the vehicle's heading edge but also enhances the accuracy of heading direction determination.

In a similar vein, OHDet advances this approach by methodically processing the region of interest (RoI), effectively classifying the four bounding box edges into distinct quadrants. One edge, specifically perpendicular to the direction of the vehicle's head, is selected as the definitive marker for the heading's orientation. The midpoint of this edge is then astutely employed to represent the heading direction, as showcased in "8+1" in Fig. 3. This methodology, when

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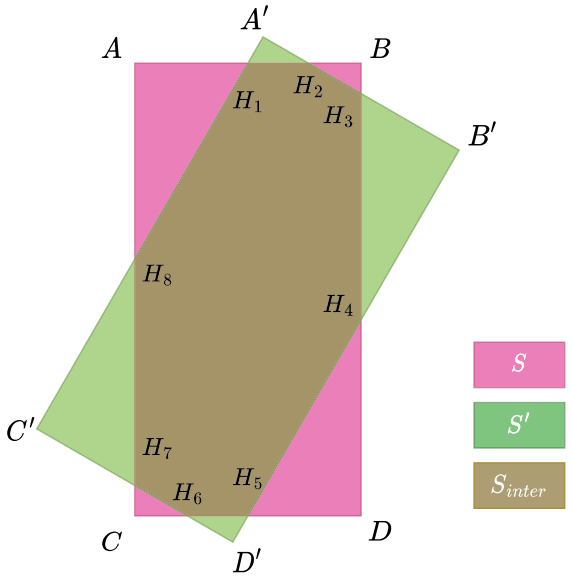


Fig. 4. Rotated-IoU method, where the changing rotational angles of the actual and prediction boxes lead to a variable number of vertices H_i, up to eight. S denotes the area within the actual box but outside the prediction box, while S' represents the area inside the prediction box but outside the actual box. These areas, S and S', are derived from a combination of up to four polygons.

applied to the five-parameter representation, is referred to as "5+1," exemplifying a refined approach to heading direction prediction in vehicle detection.

B. IoU Loss

1) IoU Loss for HBB: In the domain of horizontal object detection, Ln-norm loss and IoU are pivotal for representing HBB regression loss. IoU, especially, stands as the more favored approach. In a bid to refine traditional IoU, innovations like GIoU [30] have introduced distance metrics, while DIoU [31] extends this concept to encompass multiple dimensions, such as height, width, and center coordinates. This expansion offers a thorough portrayal of the spatial dynamics between predicted and actual boxes. CIoU [31] advances upon DIoU by integrating aspect ratio constraints, significantly bolstering the robustness of IoU. However, when it comes to the precise detection of objects regardless of their orientation, HBB proves to be limited.

2) IoU Loss for OBB: The traditional approach for rotated boxes is the Rotated-IoU, as depicted in Fig. 4. Here, (A, B, C, D) and (A', B', C', D') represent the vertices of the actual and prediction boxes. The method begins by identifying the intersection points H_1–H_8 between the two rectangles, followed by determining the position of H_i either inside or outside the intersecting area. Based on H_i's locations, the areas of various polygons and S and S' are calculated. Next, the actual intersecting area, S_{inter}, is computed through a series of additive and subtractive operations, culminating in the IoU being calculated as the ratio of S_{inter} to the union area of the two rectangles. However, this method encounters a limitation during model training, being non-differentiable at the prediction box boundaries, thus relegating its use primarily to the validation phase rather than as a loss function.

In pursuit of an IoU loss that accurately captures the characteristics of rotational boxes and simultaneously enhances model prediction during training, researchers have made significant strides. Innovations include the Smooth IoU Loss [5], which integrates an IoU constant factor with the $L1$ smoothing function, addressing abrupt increases in loss; the PIoU Loss [32] that employs approximate pixel area calculations; GWD [33], which calculates the Wasserstein distance between Gaussian distributions; and the refined KLD Loss [34]. SkewIoU [7] adapts IoU calculations for OBB with different aspect ratios.

Nevertheless, in the realm of OHD, existing IoU computation methods do not incorporate heading direction information, thus failing to depict predictive inaccuracies in heading orientation.

III. Proposed Method

A. Baseline Framework

In the vast landscape of object detection, methodologies are primarily bifurcated into single-stage and two-stage algorithms. The two-stage variants are distinguished by their ability to generate an extensive array of region proposals, culminating in superior accuracy. Single-stage algorithms, epitomized by the YOLO series, are lauded for their expeditious detection speeds. YOLOv6 [23], a paradigm of the single-stage approach, is meticulously chosen for its inherent balance of swiftness and precision, making it particularly conducive to industrial applications demanding real-time detection and control capabilities. The Baseline framework, as depicted in Fig. 5, is a symphony of three integral segments: backbone, neck, and head. The backbone emerges as the convolutional nerve center for feature extraction, the neck adeptly fuses these extracted features, and the head, leveraging an innovative anchor-free methodology [22], [35], orchestrates the nuanced task of target prediction. Within this framework, a new branch, indicated in red in Fig. 5, is added to predict the angle theta. This branch employs C_θ classifications to categorize an entire circumference into each angle for representation within the Rep-PAN [23] architecture, as depicted in the middle part of figure.

B. Circular Annotation Method

As discussed in Section II-A, current annotations for the heading angle extract only partial information from rotational box annotations. It is clear that the heading angle does not influence the rotational box annotations. Consequently, the annotation methods mentioned previously for OHD impede the effective coupling of the heading direction angle with rotational box information.

The CAM, devised to surmount this challenge, employs five parameters encompassing (x_c, y_c, w, h, and θ) for the rotated box. Here, θ is designated as the rotational angle of the heading direction, spanning a range from $[0°, 360°)$. As illustrated in Fig. 5(a), CAM takes the object's center as the origin, with the clockwise angle θ between the heading direction and true north serving as the marker of the target's

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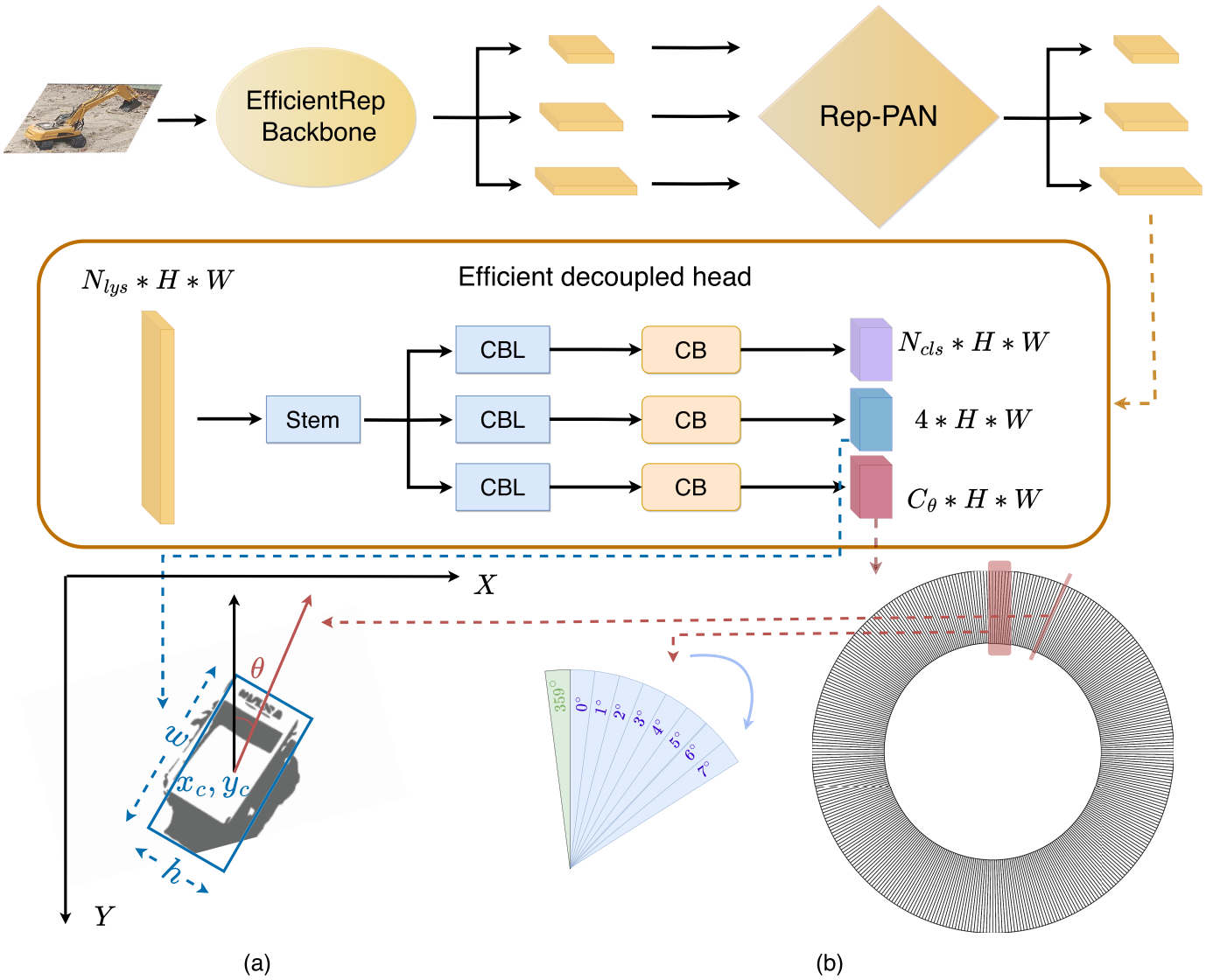


Fig. 5. Baseline framework architecture of the HDDet method. This network outputs three efficient decoupled layers. N_{lys}, H, and W represent the input sizes for the three heads: [128, 100, 100], [256, 50, 50], and [512, 25, 25]. The CBL and Stem modules consist of a convolution layer, batch norm, and SiLU. The CB module comprises $1 * 1$ convolutional kernels. N_{cls} denotes the number of classes in the dataset. As depicted in (a), The quartet value 4 encapsulates the bounding box's central coordinates (x_c, y_c), width w, and height h, respectively. Moreover, as depicted in (b), C_θ indicates the number of parameters required for angle prediction, where C_θ representation is represented using the FullCSL method.

heading direction angle. This alignment, combined with horizontal box parameters, manifests as a rotational box, with θ measured in degree. This nuanced annotation ensures a unique angle corresponds to every possible heading direction, substantially elevating accuracy. Through this mechanism, the angles of heading direction and the rotational box are coupled. CAM's adaptability ensures that predictions of heading direction remain unaffected by the prediction box's aspect ratio, enabling precise representation for OHD.

To transition from the existing "5+1" annotation to CAM, Algorithm 1 was formulated. First, the heading direction vector V and the true north vector V' are calculated by center point (x_c, y_c) and heading point (x_h, y_h). The algorithm then calculates the counterclockwise rotational angle of V relative to V' using the arctangent function, subsequently translating it into a clockwise orientation.

For "8" and "8+1" annotations, where vertex sequencing is predefined, the heading position is demarcated by the midpoint of the line connecting the first two vertices. Existing algorithms transition from eight-point to "5+1" annotations, eventually evolving into CAM annotation via Algorithm 1. This transformative approach also facilitates the conversion of datasets like DOTA's original eight-parameter annotations into circular annotations, streamlining dataset preprocessing.

C. FullCSL Angular Representation

To precisely detect the target with CAM, the rotational angle of heading θ is needed to accurately predicted. Although regression can provide continuous angle predictions, more accurately describing the rotation of an object, it has issues at boundary conditions, leading to sudden increases in loss and affecting model accuracy. Thus, performing regression

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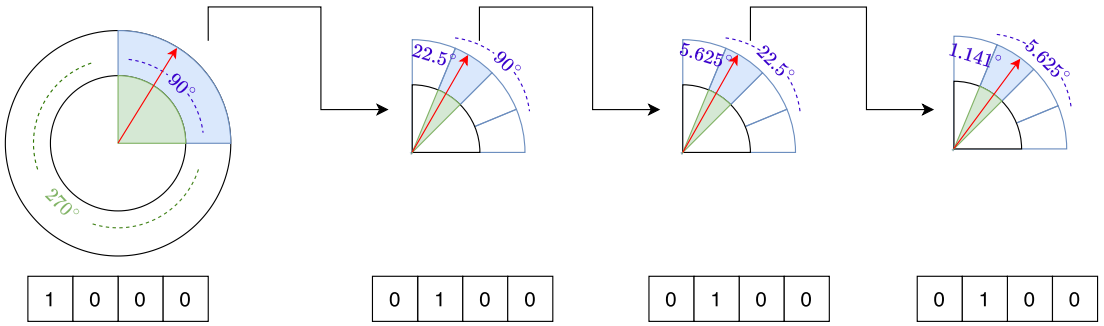


Fig. 6. One example of MDAE: calculating 30° for $K = 4$, $N = 4$. The computation involves sequential operations: $0 + 22.5 + 5.625 + 1.40625 = 29.53125$.

Algorithm 1 Point Decode to Angle

Input : Bounding box center x_c, y_c. Width w. Height h. Heading direction point x_h, y_h.
Output: Heading direction angle

Function
  V_x,V_y=(x_h-x_c,y_h-y_c);
  V'_x,V'_y= (0,1);
  $angle$ = atan2(V_x * V'_y - V_y * V'_x, V'_x * V_x + V'_y * V_y);
  $angle$ = 360° - $angle$ % 360°;
  return $angle$;

prediction over a large interval like 360° can be challenging. Inspired by CSL [11], we propose FullCSL, which consists of two steps: the discretization of angle theta and the CSL calculation. In the first step, the continuous variable θ in CAM is approximated using a classification approach. As shown in Fig. 5(b), the circular interval is divided into 360 categories, starting from 0 degree, with each category covering a range of 1 degree. Hence, the continuous angle value can be represented by a discrete classification index, which is predicted by our angle prediction branch.

In the second step, the value of FullCSL for any angle is calculated. The expression for FullCSL is shown in (1). x_θ signifies each individual degree within the full scope of 360 degree. The smoothing parameter σ is set to 6. This culminates in our angular labels. According to CSL, which addresses the issues of EoE and PoA, it is evident that the values calculated by FullCSL are better suited for depicting the relationship between prediction angles and targets. Therefore, these values will be used to measure the angular loss in the loss function

$$\text{FullCSL}(\theta) = \left\{ e^{-\frac{(x_\theta-\theta)^2}{2\sigma^2}}, \right\}_{x_\theta=0}^{359}. \tag{1}$$

D. Multidimensional Angle Encoding

The MDAE method addresses the critical challenge posed by the voluminous prediction tensor, i.e., $360*H*W$ of FullCSL. This method streamlines the predictive process by deploying a tiered angle stratification strategy. MDAE encodes an angle by N distinct scales and K equal segments in each scale. This stratification not only significantly reduces the dimensionality of angle prediction but also preserves the essential accuracy and periodicity inherent in angular measurements.

As depicted in Fig. 6, in the first scale, MDAE partitions the whole 360° range into K segments, each covering the angle range of $360°/K$. Then the second scale is the $360°/K$ angle range. If the heading angle falls on the kth segment of the first scale, we will further partition the angle range $[360° * (k-1), 360° * k]$ into K new segments. Each new segments cover the angle range of $360/K^2$. We keep repeating the above partition until the Nth scale. Hence, any angle θ can be approximated with (2), where k_n is the index of the angle segment of the nth scale on which θ falls.

According to (2), the encoding and decoding processes are summarized in (3). The encoding process concatenates N Onehot labels of length K, with the k_nth bit in each is 1. In the decoding process, $\boldsymbol{x}$ is the predicted value of the angle prediction branch, whose length is $N*K$ for each anchor point. $\boldsymbol{x}_{K(n-1):Kn}$ denotes the predicted sequence of the nth scale, which are calculated by the sigmoid activation and selected the index of the max value by function max_index(·) as the prediction of k_n. Finally, the predicted angle theta can be calculated by the following equation:

$$\theta \approx \sum_{n=1}^{N} \frac{360}{K^n}(k_n - 1) \tag{2}$$

where $k_n \in \{1, 2, \ldots, K\}$

$$\begin{aligned} \textbf{encoding}: & \left\{ 0, 0, \ldots, \underset{\uparrow}{1}, 0, \ldots \right\}_{c_n}^{N}{}_{n=1} \\ \textbf{decoding}: & \ \widetilde{\theta} = \sum_{n=1}^{N} \frac{360}{K^n} \text{max_index} \\ & \left(\text{sigmoid}\left(\boldsymbol{x}_{K(n-1):Kn}\right)\right). \end{aligned} \tag{3}$$

The precision and maximum error of the MDAE method are intricately showcased in Table I. Here, AR denoting the total angular range. The maximum error of MDAE is deduced as $180/K^N$, while the average error stands at $90/K^N$. Notably, with parameters set at $N = 6, K = 6$, MDAE achieves a minimal systematic error of 0.0039. This highlights MDAE's capability to maintain exceptional angular accuracy and periodicity, all while streamlining the model's head layer.

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TABLE I
COMPARATIVE OF MAXIMUM ERRORS IN DIFFERENT ANGULAR REPRESENTATION METHODS

Method	AR	C_θ	Angular Periodicity	Max Error
CSL	180	180	Y	0.5
DCL	256	8	N	0.3515625
MDAE(4,4)	360	16	Y	0.703
MDAE(6,6)	360	36	Y	0.0039

E. CosWeight

IoU, functioning as a comprehensive metric and loss function, seamlessly integrates all elements of the predicted bounding box, thereby mirroring the disparity between the predicted and actual boxes. Recognizing that the HBB IoU omits critical angular details, and the Rotated-IoU for the OBB is hindered by its non-differentiability despite including angular information, we introduced CosWeight. This innovative approach, which infuses heading angle and Rotated-IoU losses into the loss function, is directly multiplied with the HBB's IoU. This fusion not only mimics the characteristics of Rotated-IoU but also incorporates the dynamic changes in heading angle, thus embedding the essential data needed by the OHD within the loss computation framework.

In actuality, the variation in Rotated-IoU is influenced by both the rotational angle and the aspect ratio M. Addressing this, our analysis encompassed three distinct aspect ratios [as illustrated in Fig. 7(a)], examining the IoU's fluctuation with the rotational angle in these specific contexts. As depicted by the blue line in Fig. 7(b), the Rotated-IoU's angle difference exhibits a periodicity of 180°, resulting in its symmetry around the 180-degree axis.

CosWeight employs a nuanced composite function of cosine, $f(\cos(2\theta), M)$, adeptly approximating Rotated-IoU while capturing the essence of IoU information and avoiding the pitfalls of non-differentiability. Given that Rotated-IoU operates on a 180° cycle and the actual angular disparity 2θ spans a 360° range, we adjusted the parameter by halving 2θ to θ. This adjustment extends its cycle to encompass 360°, aligning perfectly with the heading angle requirements. The detailed design process is as follows.

To commence, Monte Carlo simulations are harnessed for the polynomial approximation of the $f(\cos(2\theta), M)$ function, where $M = \max(w, h)/\min(w, h)$ delineates the actual bounding box's configuration. Dynamic time warping (DTW) is adeptly utilized to evaluate the divergence between two curves [referenced in (4)], as shown at the bottom of the next page, adhering to the constraints defined in (4). S_1 and S_2 correspondingly symbolize $f(\cos(2\theta), M)$ and Rotated-IoU(θ, M_i), for $i \in [1, 2, 3]$

Following this, we employed a Gaussian smoothing function to refine $f(\cos(2\theta))$, yielding (5) and (6). Notably, φ reaches its maximum value of 1 when $M = 1$, and displays symmetry around this value, satisfying the stipulations for M

$$\gamma = \frac{0.22}{\varphi} cos(2\theta) + 0.22e^{0.48\varphi} + 0.32 \tag{5}$$

$$\varphi = e^{-\left(\frac{w}{h}-1\right)^2} + e^{-\left(\frac{h}{w}-1\right)^2} \tag{6}$$

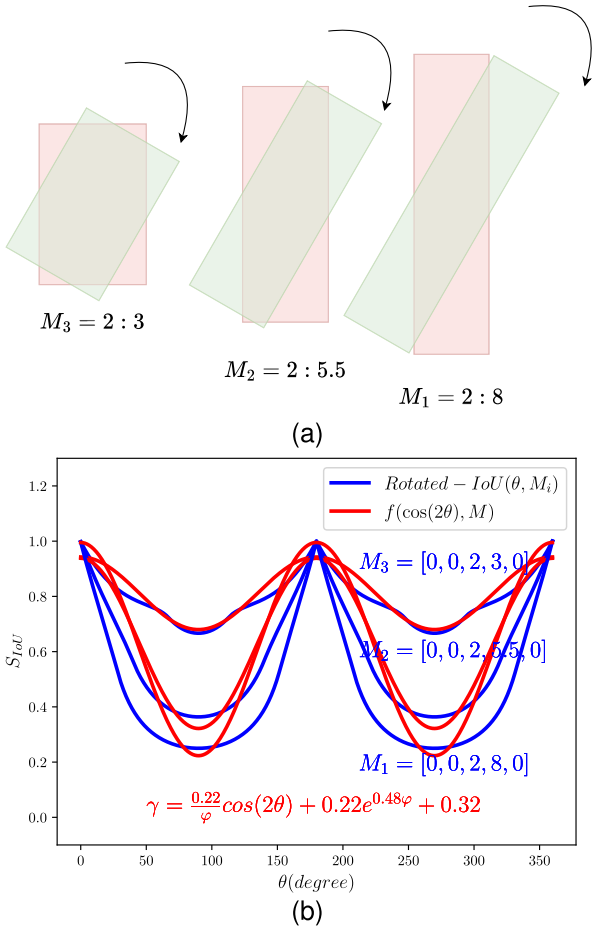


Fig. 7. Matrices M_1, M_2, and M_3 act as exemplars of predicted bounding boxes within the dataset. θ signifies the 360° rotational trajectory of bounding box M around its central point. The blue curve delineates the IoU's variation with angle, while the red curve depicts the corresponding fit graph. (a) Three typical ratios of predicted bounding boxes. (b) Graphical representation of CosWeight fitting.

$$\begin{aligned} \text{CosWeight}(\theta, M) &= 1 - \gamma' \\ &= 0.68 - \frac{0.22}{\varphi} cos(\theta) - 0.22e^{0.48\varphi}. \end{aligned} \tag{7}$$

At this juncture, as the red line in Fig. 7(b) demonstrates, the DTW distance is a mere 8. γ acts as a non-essential derivative, obviating the need for gradient computations. Although the formula contains numbers that may appear inexact, they suffice in gauging the proximity between diverse positive samples and the actual bounding box. Ultimately, concerning the heading direction in circumference, we adjusted the parameter 2θ to be θ to achieve a 360° cycle. The definitive CosWeight is specified in (7), with its depiction in Fig. 8(b). As Fig. 8(a) reveals, when the red and purple boxes display similar Rotated-IoU values but a substantial difference in heading angles, there is a noticeable disparity in their respective CosWeight values.

F. Loss Function

In our advanced OHD approach, the prediction box is defined using six parameters: (x_c, y_c, w, h, c, and θ). These include (x_c, y_c) for the positional offsets of the center, (w, h) for the box's dimensions, c for classifying the object category,

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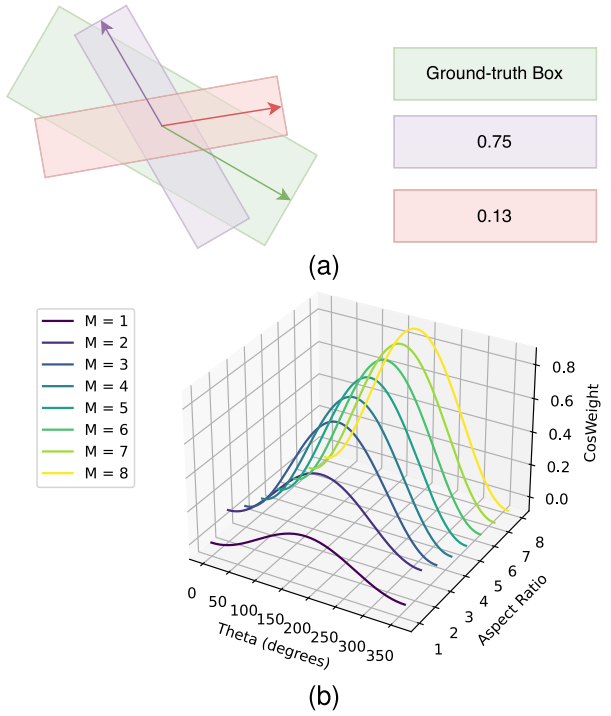


Fig. 8. (a) CosWeight values at two distinct angles, highlighting a significant increase in loss when the predicted angle sharply diverges from the true orientation of the box. (b) Graphical representation of CosWeight, indicating minimal error as the heading angle nears 0° or 360°, with a pronounced increase in loss around 180°. M denotes the aspect ratios of various bounding boxes, with eight selected ratios from 1 to 8 showcased here.

and θ for the crucial rotational angle. Embracing the varifocal loss (VFL) [36] for classification, we optimize the model's ability to discern and categorize objects with precision. The choice to exclude object loss from our anchor-free configuration is informed by its limited impact on performance enhancement [23]. The box regression loss is calculated using the smooth $L1$ loss method, complemented by the binary cross-entropy loss for angle estimation. The composite loss function is elegantly formulated in the following equation:

$$\begin{aligned} L_{\text{total}} &= \frac{1}{N}\left(\lambda_1 L_{\text{cls}} + \lambda_2 L_{\text{box}} + \lambda_3 L_{\text{ang}}\right) \\ &= \lambda_1 \frac{1}{N} L_{\text{cls}}\left(c_t, c_p\right) \\ &\quad + \lambda_2 \frac{1}{N} L_{\text{box}}(T_{\text{box}}, P_{\text{box}}) \\ &\quad + \lambda_3 \frac{1}{N} L_{\text{ang}}\left(\theta_t, \theta_p\right). \end{aligned} \tag{8}$$

Detailed definitions of L_{cls}, L_{box}, and L_{ang} are as follows:

$$\begin{aligned} L_{\text{cls}}\left(c_t, c_p\right) &= \text{VarifocalLoss}\left(c_t, c_p\right) \\ L_{\text{box}}(T_{\text{box}}, P_{\text{box}}) &= \text{CosWeight}\left(\theta_{\text{t}} - \theta_{\text{p}}, \text{w}_{\text{t}}, \text{h}_{\text{t}}\right) \\ &\quad * L_{\text{IoU}}(T_{\text{box}}, P_{\text{box}}) \\ L_{\text{ang}} &= \text{BCE}\left(\theta_t, \theta_p\right). \end{aligned} \tag{9}$$

The parameters for the actual box, $(x_t, y_t, w_t, h_t, c_t, \theta_t)$, and those for the predictive box, $(x_p, y_p, w_p, h_p, c_p, \theta_p)$, are meticulously crafted. $(T_{\text{box}}, P_{\text{box}})$ signify the comparison between actual and predicted boxes, respectively. L_{IoU} is using GIoU by default. λ_i are strategically set hyperparameters, balancing the influence of each loss component within the total loss equation.

IV. Experiments

To verify the performance of our HDDet, an experimental setup was crafted on a high-end server equipped with NVIDIA GeForce RTX 4090 GPUs, operating under Ubuntu 22.04.3 LTS and using PyTorch version 2.0.1. The underlying network, modified from the YOLOv6 framework, was primed for optimal performance. The training regimen was anchored by the EffiDeHead, an anchor-free detector, synergized with the AdamW optimizer. This setup was fine-tuned with a learning rate of 2.5e-4, weight decay at 3.6e-4, and momentum finely balanced at 0.843, ensuring a robust and efficient training process.

A. Datasets

1) Our Datasets: In the realm of 2-D engineered vehicle object detection, our dataset stands out as a pioneer, primarily focusing on the directional aspect of vehicle heading. It encompasses 789 images with different viewing angles, covering three distinct vehicle types: dumpers, bulldozers, and navvies, cumulatively accounting for 2201 instances. These images, resonating with 1024P clarity, depict scaled models of real-world construction vehicles, remotely controlled to simulate authentic operational scenarios. The diverse shooting environments span grassy terrains to sandy and concrete surfaces, under varying weather conditions from sunny to foggy, captured at distances extending from 0 to 12 m. The dataset is thoughtfully divided, comprising 658 training and 131 validation images, each meticulously annotated with coordinates (x_c, y_c, w, h, θ), where θ spans a comprehensive range of $[0°, 360°)$.

$$\begin{aligned} \text{DTW}(i, j) &= \begin{cases} 0, & i = j = 0 \\ +\infty, & i = 0 \text{ or } j = 0 \\ |x_i - y_j| + \min(\text{DTW}(i-1, j), & \\ \text{DTW}(i, j-1), \text{DTW}(i-1, j-1)) & \text{otherwise} \end{cases} \\ &\min\left(\text{DTW}_{m_1}(S_1, S_2), \text{DTW}_{m_2}(S_1, S_2), \text{DTW}_{m_3}(S_1, S_2)\right) \\ &\approx \frac{1}{N}\sum_{i=1}^{N}\left(\text{DTW}^i_{m_1}(S_1, S_2), \text{DTW}^i_{m_2}(S_1, S_2), \text{DTW}^i_{m_3}(S_1, S_2)\right). \end{aligned} \tag{4}$$

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2) HRSC2016 [37]: The HRSC2016 dataset, a treasure trove for ship detection in remote sensing, features imagery from six renowned harbors, totaling 2976 targets. It is categorically split into 436 training, 181 validation, and 444 test images. Each image meticulously marks the coordinates of the bounding box's four corners and the ship's heading. We have adeptly converted these heading coordinates into angular information and uniformly resized the images to 800×800, establishing a standardized training and evaluation procedure.

3) DOTA [38]: The DOTA dataset stands as a testament to complexity and quality in large-scale remote sensing imagery, comprising 2806 images, some extending to the vast dimensions of 4000×4000. It showcases an impressive collection of 188 282 bounding boxes across 15 categories. Focusing on categories like plane (PL), small vehicle (SV), large vehicle (LV), ship (SH), harbor (HA), and helicopter (HC), we have transformed their bounding boxes, annotated with ordered coordinates, into angular data equipped with heading direction. These images were then segmented into multiple 1024×1024 units with a 200-pixel overlap, subsequently resized to 800×800, thereby facilitating a cohesive and effective training and evaluation regimen.

4) OHD-SJTU [10]: OHD-SJTU emerges as an exemplary dataset, publicly available and ingeniously tailored for advancements in rotation detection and object heading direction delineation. This dataset bifurcates into OHD-SJTU-L and OHD-SJTU-S, each offering unique insights. The OHD-SJTU-S portion, enriched with high-resolution imagery from Google Earth, features 43 extensive scene images, boasting up to $16\,000 \times 16\,000$ pixels. It encapsulates two pivotal categories—ships and planes—with a collective tally of 4125 instances. The expanded OHD-SJTU-L segment further enriches the dataset, adding categories like small and large vehicles, harbors, and helicopters, all sourced from DOTA and enhanced with directional heading annotations.

OHD-SJTU-L, with its substantial collection of 113 435 instances, served as a cornerstone in our experimental analysis. For a thorough examination, we methodically segmented the images into 600×600 sub-images, maintaining a 150-pixel overlap, and subsequently scaled them to 800×800, setting the stage for detailed analysis.

B. Evaluation Metrics and Protocols

1) Evaluation for Bounding Box: In the evolving landscape of object detection, the shift from HBBs to OBBs has solidified mAP as a paramount metric. It is meticulously calculated based on IoU values, adhering to the VOC07 standard. Our evaluation extended beyond the conventional, gauging its computational demand in floating point of operations (FLOPs), thus providing a comprehensive assessment of its performance.

2) Evaluation for Heading Detection: In the specialized field of OHD, mAP alone does not suffice for a comprehensive assessment. Particularly in industrial applications like remote operations, the precision of heading direction detection often eclipses the significance of mAP. When faced with scenarios of similar mAP values, it is imperative for OHD evaluations to delve deeper, examining and quantifying accuracy in these critical aspects

$$\text{headacc} = \frac{1}{N}\sum_{i=1}^{N} \text{count}\left(\text{distance}\left(\theta_{i_t}, \theta_{i_p}\right) \le r\right) \tag{10}$$

$$\text{headloss} = \frac{1}{N}\sum_{i=1}^{N} \text{distance}\left(\theta_{i_t}, \theta_{i_p}\right). \tag{11}$$

In our analytical framework, N is the parameter representing the quantity of predicted boxes that exceed a predetermined confidence threshold. The pair $(\theta_{i_t}, \theta_{i_p})$ represents the actual and predicted angles for the ith box, while the function distance$(\cdot)$ measures the extent of angular deviation within a circular construct. And count$(\cdot)$ is used to record the count.

Particularly for objects like "plane," characterized by a smaller aspect ratio, there is a notable challenge: despite the prediction box being accurately placed, the direction of the heading often deviates. This phenomenon is clearly illustrated in Fig. 2(a), (c), and (d). Under these circumstances, the focus shifts to the precision of the angle. 'headacc' (heading accuracy) emerges as a vital metric here, assessing the ratio of samples whose predicted angle deviation falls within a specific range r, commonly set at 5 degree.

"Headloss," conversely, is the metric that quantifies the average error in angular predictions. It becomes particularly critical for objects with larger aspect ratios, such as "harbor," where predictions of the angular direction might be entirely inverted, as seen in Fig. 2(b). Moreover, in scenarios where datasets exhibit lower complexity and "headacc" fails to adequately reflect model effectiveness, 'headloss' proves to be a more suitable and discerning metric. In the realm of industrial applications, the headacc metric serves as an efficient tool for quantifying the degree of oscillation in a vehicle's heading. Concurrently, headloss is employed to measure the probability of encountering reverse predictions concerning the direction of a vehicle's heading.

This rigorous evaluation process entails a thorough analysis of each positive sample following every batch prediction. The task involves pinpointing the actual box with the highest IoU corresponding to each positive sample and then calculating both "headloss" and "headacc" for these matched boxes. The culmination of this process is the averaging of these metrics across all boxes, providing a comprehensive and nuanced understanding of the model's overall performance.

C. Ablation Study

1) Comparison of Different Scales and Segments: In our methodical exploration, the number of scales and segments, N and K, emerges as pivotal hyperparameters, each playing a distinct role in model optimization. The choice of N is crucial, dictating the initial scale's breadth, thereby directly influencing the rapidity and efficiency of the model's convergence. Contrastingly, K steers the granularity post-scale, a determinant factor in ensuring the model's fitting accuracy. Our experimentation spanned various permutations of N and K, as systematically presented in Table II. The analysis involved

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TABLE II
DETAILED COMPARISON OF RESULTS UNDER DIFFERENT SCALE SETTINGS

MDAE Method	Loss Weight	mAP_{50}	Headacc	Headloss
$N = 4, K = 4$	0.05	76.58	77.58	77.58
$N = 6, K = 6$	0.05	91.44	83.68	**75.31**
$N = 6, K = 5$	0.05	93.88	83.79	79.61
$N = 6, K = 4$	0.05	93.70	83.88	80.94
$N = 7, K = 5$	0.05	59.14	70.03	85.39
$N = 8, K = 8$	0.05	51.43	81.45	86.76
$N = 6, K = 5$	0.08	94.34	**88.23**	83.93
$N = 6, K = 6$	0.08	**94.50**	86.81	**75.31**

comparing their performance impacts, refined through meticulous adjustments in the angle loss weight.

Employing the HRSC2016 dataset, our experiment leveraged the lightweight YOLOv6-N model, subjected to an intensive 200-epoch training regime on the pre-trained COCO weights. We identified the best-performing weights for a detailed comparative study. The ($N = 4$, $K = 4$) configuration exhibited underwhelming results, attributable to its coarse granularity. In stark contrast, the ($N = 8, K = 8$) setting encountered hindered learning efficiency, a consequence of its overly segmented approach. We adopt ($N = 6$, $K = 6$) as the optimal configuration for MDAE, striking a balance between segmentation depth and learning proficiency.

2) Comparison of Different Angle Representations: The DOTA dataset, not initially tailored for heading direction detection, presented challenges. Our selection criteria for the comparative study focused on models achieving the highest mAP within the initial 200 epochs. We evaluated MGAR [39], an esteemed OBB angle representation method, extended to encompass a $[0°, 360°)$ angle prediction range. This evaluation included MGAR, FullCSL, and MDAE, all utilizing classification-based approaches for angle representation. FullCSL employed a Gaussian smoothing parameter $\sigma = 6$, while MDAE operated under parameters ($K = 6$, $N = 6$). C_θ signifies the output dimension of the angle layer. Conducted on the DOTA dataset under uniform hyperparameter conditions, this comparison yielded insightful results, as detailed in Table III.

The analysis revealed that while OBB methods secure high mAP, they falter in accurately detecting heading direction. Conversely, both FullCSL and MDAE excel in heading direction prediction, managing to maintain robust mAP performance. FullCSL, characterized by a substantial 360-dimensional angle tensor, necessitates extended training durations. Its Gaussian smoothing approach slightly compromises accuracy in detecting near-square objects, yet it exhibits a marked advantage in recognizing objects with more pronounced aspect ratios. MDAE, by contrast, combines both efficacy and expedited convergence for faster and more effective short-term training, offering an advantageous blend of these qualities.

3) Impact of CosWeight: In the context of utilizing a standard OBB detector, it is frequently observed that object heading predictions are inverted during the initial stages of model training, as exemplified in Fig. 9. The CosWeight approach effectively navigates the early learning process of heading angles, ultimately resulting in an increase in OHD accuracy. CosWeight contains both the Rotated-IoU and heading angle information, offering a robust measure of the proximity between positive samples and the ground truth. We tested this methodology across four diverse datasets: HRSC2016, DOTA, UCAS(plane), and Our dataset. These tests unequivocally illustrate CosWeight's effectiveness in refining angle predictions, as substantiated in Table IV.

Especially in complex datasets like DOTA, the advantages of CosWeight are unmistakably prominent, evidenced by a substantial increase in heading accuracy of up to 12.47%. Across all the datasets, there was a discernible decrease in

TABLE III
COMPARATIVE ANALYSIS OF DIFFERENT ANGULAR REPRESENTATIONS IN THE DOTA DATASET

Angle Representation	C_θ	mAP_{50}	Headacc	Headloss	Gflops
MGAR	11	**72.54**	45.63	71.54	**243.29**
FullCSL	360	71.31	**84.35**	34.79	245.35
MDAE	36	71.47	80.27	**30.08**	243.9
MDAE (600 epochs)	36	**78.34**	66.79	**27.87**	243.9

Note: Due to the differences in the types of large-vehicles and the sequence of starting points in the DOTA dataset, some results are affected. However, with prolonged training, the MDAE can still attain predictive capabilities close to those of an OBB detector.

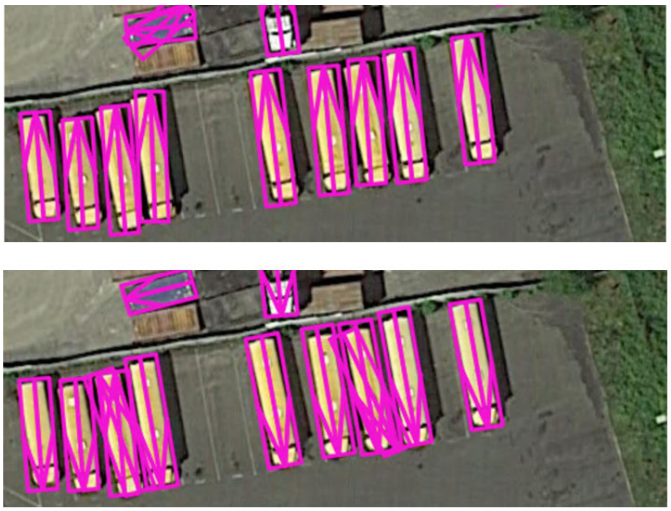


Fig. 9. Initial stages of model training often witness reversed predictions in heading direction (illustrated in the upper subplot). The integration of MDAE with CosWeight proficiently addresses this challenge (as seen in the lower subplot).

TABLE IV
IMPACT OF COSWEIGHT IN DIFFERENT DATASETS

Dataset	Method	mAP_{50}	Headacc	Headloss
HRSC2016	Baseline	89.06	75.28	81.69
	Baseline+CosWeight	**91.44**	**83.68**	**75.31**
DOTA	Baseline	**73.55**	67.80	37.57
	Baseline+CosWeight	71.47	**80.27**	**30.08**
UCAS(Plane)	Baseline	**91.71**	34.50	85.80
	Baseline+CosWeight	89.97	**46.95**	**74.04**
Our Datasets	Baseline	86.25	**99.56**	39.03
	Baseline+CosWeight	**87.39**	99.38	**29.15**

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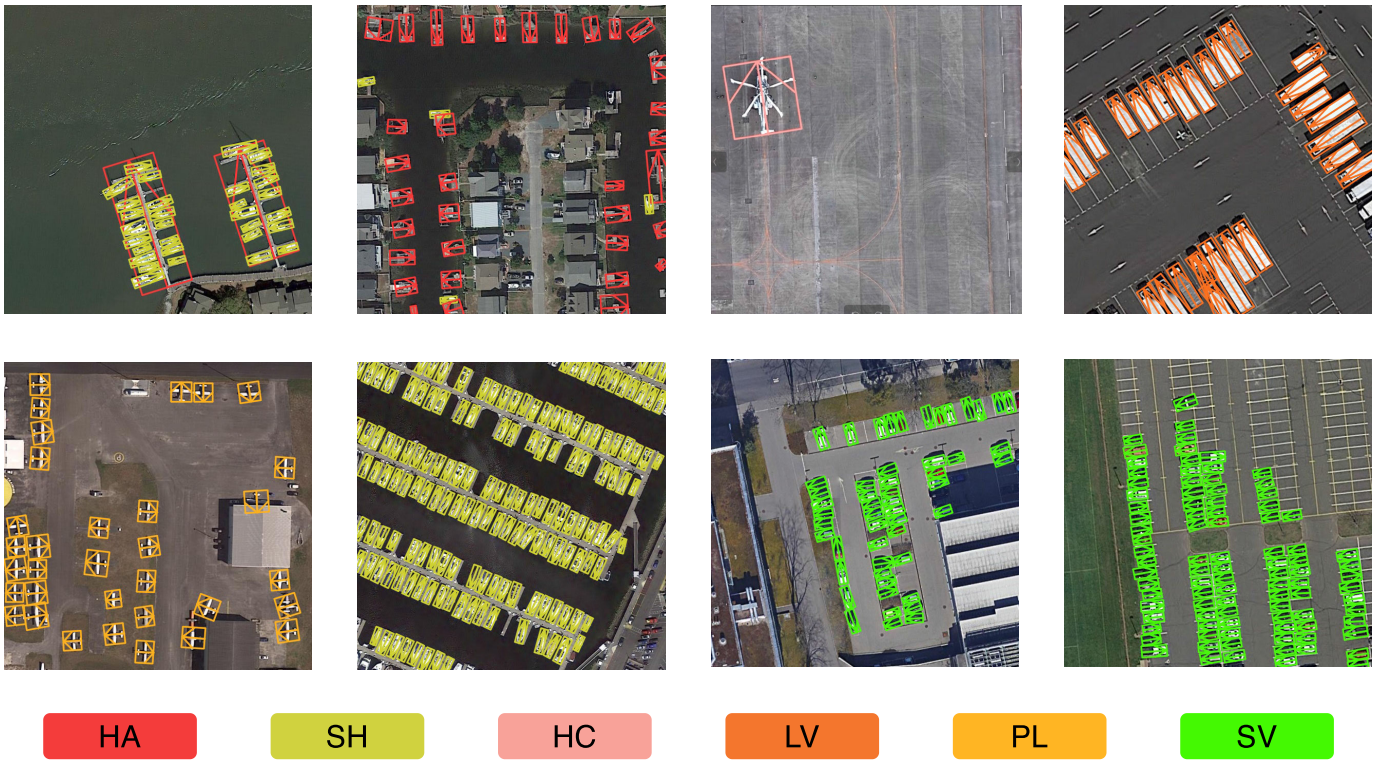


Fig. 10. Showcase of the HDDet's performance on the OHD-SJTU-L dataset.

TABLE V
COMPARISON OF AP BETWEEN OHDET AND HDDET

Method	HA	HC	LV	PL	SH	SV	mAP_{50}
OHDet(OBB)	**61.43**	33.87	**76.55**	**89.62**	85.58	48.45	65.92
OHDet(OHD)	41.29	17.53	35.32	59.93	47.57	26.59	38.04
HDDet	49.58	**38.31**	68.39	85.35	**88.97**	**70.16**	**66.81**

TABLE VI
HEADING ACCURACY COMPARISON BETWEEN OHDET AND HDDET

Method	HA	HC	LV	PL	SH	SV	mAP_{50}
OHDet(OHD)	76.66	49.06	57.79	74.43	68.39	60.15	64.41
HDDet	**98.99**	**96.24**	**93.18**	**95.63**	**94.14**	**90.74**	**94.82**

Note: The heading accuracy (headacc) metric, with a radius r of 45, approximates OHDet's four-category evaluation criterion.

heading loss. Speed tests, encompassing pre-processing, inference, and NMS time, averaged across ten trials, achieved an outstanding rate of 106 FPS on an RTX 4090, demonstrating the efficiency and effectiveness of the approach.

D. Comparison With State-of-the-Art

OHDet, heralded as a cutting-edge model in OHD, has made significant strides in OBB detection. However, its approach, characterized by an additional four-category paradigm without coupling the information of heading with OBB detection, has been noted to impede optimal heading detection performance. In stark contrast, HDDet correctly couples the information of heading and OBB by CAM, ensuring that heading direction estimation does not detrimentally impact bounding box precision. This innovative approach not only enhances the model's efficiency but also markedly improves its accuracy compared to OHDet.

An in-depth comparative evaluation between HDDet and OHDet was executed on the OHD-SJTU-L dataset. The detailed AP comparison findings are delineated in Table V. Within the realm of OBB detection, HDDet demonstrates superior performance in identifying helicopters, small vehicles, and ships, registering a 0.89% enhancement in mAP relative to OHDet. In terms of OHD, HDDet led across the board in all categories with a 28.77% improvement in mAP. The heading precision outcomes, as depicted in Table VI, reveal a remarkable ascension in overall heading prediction accuracy from 64.41% to 94.82% post the implementation of CAM. HDDet not only exceeds OHDet in mAP metrics but also maintains a lead in the precision of heading detection. Fig. 10 presents an array of predictive outcomes from the OHD-SJTU-L dataset.

E. OHD in Arbitrary Viewing Angle

Contrasting with vertical perspectives, the performance of object heading detectors at arbitrary viewing angles is profoundly reliant on their adeptness in accurately determining headings. Within our engineering vehicle dataset, the training set's limited array of vehicle orientations does not fully encompass all possible spatial headings. In an effort to bolster the model's resilience, we ventured into utilizing data augmentation to amalgamate features from diverse perspectives, as depicted in Fig. 11. Although this augmentation approach may cause some deformation of the target objects, it has notably optimized the model's overall functionality.

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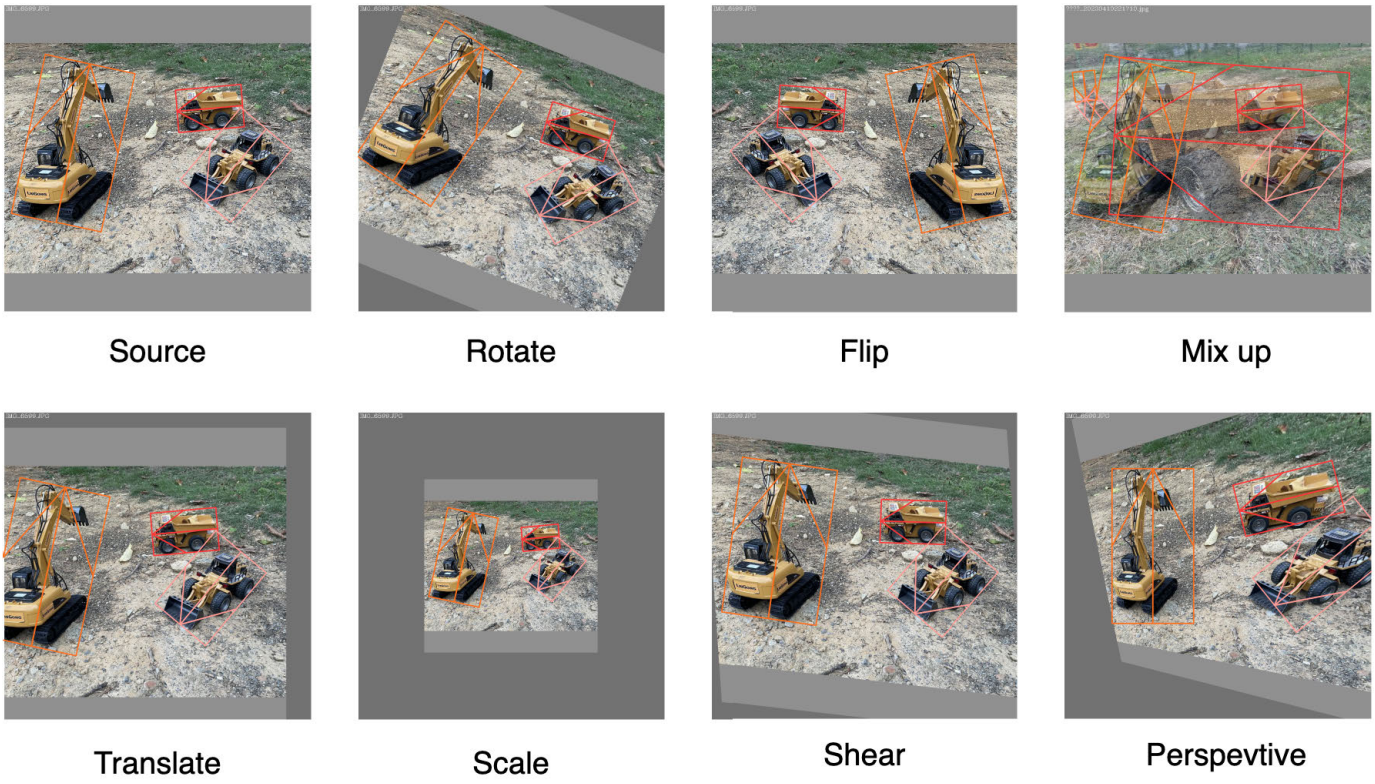


Fig. 11. Comparison of different data augmentation on our dataset.

TABLE VII
COMPARISON OF DIFFERENT DATA AUGMENTS

Augment Method	None	Rotate(45°)	Flip(0.5)	Mix up(0.5)	Translate(0.1)	Scale(0.5)	Shear(0.1)	Perspective	**Shear+Scale+Translate**
mAP_{50}	70.81	83.13	80.00	80.34	**86.67**	84.60	**87.02**	83.96	**88.32**
$mAP_{50:95}$	31.35	35.75	33.56	33.59	**40.15**	39.67	**40.82**	39.87	**40.92**
headacc	**100.00**	98.87	98.89	**99.60**	99.58	98.59	99.05	99.55	99.15
headloss	42.37	49.24	53.20	43.81	39.82	**34.83**	**35.77**	38.65	38.20

Note: The top two results for each assessment metric are highlighted here. After conducting training sessions with OHDet on our dataset, utilizing the default settings, the model attained an average heading accuracy of 43.56%. Delving into specifics, accuracies for different vehicle types such as dumpers, bulldozers, and navvies were recorded at 42.00%, 43.59%, and 45.10%, respectively. These findings distinctly underscore that OHDet's capability in identifying objects from varied angles is somewhat limited.

Referencing Table VII, prior to implementing data augmentation strategies, the model showcased a notable proficiency in head recognition, yet its mAP was constrained to a mere 70.81%. This limitation highlighted an inadequacy in capturing the objects' diverse textural features from multiple perspectives. Introduction of any data augmentation technique fostered a marked enhancement in mAP when compared to the control group. Particularly, the shear and translate methods contributed to mAP increments of 15.86% and 16.21%, respectively, over the control group. Moreover, the synergy of shear and scale methods improved the accuracy of heading detection. Despite the deformations incurred on the target object due to the image perspective processing effect, the method yields a significant enhancement in heading accuracy. As a result, shear, scale, and translate were selected as the optimal data augmentation approaches.

V. CONCLUSION

In this study, we introduced the HDDet method, a novel solution for accurately detecting object heading directions in target detection tasks. This method is meticulously crafted, encompassing innovative approaches in data annotation, angle representation, and loss function development. Remarkably, it performs efficiently in backpropagation learning across varying object aspect ratios. Our experiments on a specialized custom-engineered vehicle dataset validate HDDet's exceptional transfer learning ability, adeptly catering to heading direction detection from both aerial and horizontal perspectives. Comprehensive evaluations on diverse public datasets confirm our method's superior proficiency in OBB detection and its exemplary state-of-the-art performance in heading direction detection. These findings not only underscore the efficacy of HDDet but also open new avenues for future research in this domain.

3-D object detection has swiftly risen to prominence in recent research landscapes. The potential of multiperspective heading detection in augmenting both visual-centric and multimodal methodologies within the 3-D object detection arena is substantial. This approach is poised as a rapid and effective solution for extracting critical vehicle heading information

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and 3-D proposal boxes in 2-D detection, marking it as a prospective research frontier [40]. On the flip side, while CosWeight markedly boosts the precision of heading predictions, its influence on the mAP warrants attention. Thus, the CosWeight method embodies considerable scope for further refinement and advancements.

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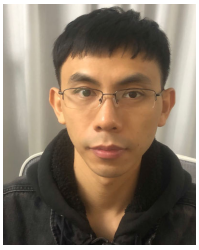
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